Fill in the table, with attention

Same rule. This time most of the table is switched off.

The reference is **TATTATGCA**. The read is **TATTTGCA**. One letter is missing from the read. We first slid the read along the reference to see where the letters line up best, and kept only the cells near that place. **Black cells are switched off: leave them empty, and count them as 0.**

The rule for one cell

Look at three neighbours and work out three numbers:

from the **upper-left** cell:	same letters: add 2	different letters: subtract 1
from the cell **above**:	subtract 2	
from the cell to the **left**:	subtract 2	

Write the **biggest** of the three. If the biggest is below 0, write **0**.

The border, in italics, is all zeros. Two cells are done for you:

- T against T: upper-left is 0, same letters, so $0 + 2 = 2$. Above gives -2, the cell to the left is black and counts as 0, so left gives -2. Write 2.
- T against A: upper-left gives $0 - 1 = -1$. Above gives -2, left gives $2 - 2 = 0$. Write 0.

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	2	0							
A	*0*									
T	*0*									
T	*0*									
T	*0*									
G	*0*									
C	*0*									
A	*0*									

When the white cells are full

1. What is the biggest number in the table, and where is it?
2. Start there. Ask which neighbour that number came from, and step to it. Repeat until you reach a 0. Draw the path.
3. One step of your path goes **left** instead of diagonally. That is the missing letter. Which one is it?
4. How many cells did you fill in? How many would the whole table have?

Answer

		T	**A**	**T**	**T**	**A**	**T**	**G**	**C**	**A**
	0	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*	*0*
T	*0*	[2]	0	2						
A	*0*		[4]	2	1					
T	*0*			[6]	4	2				
T	*0*				[8]	[6]	4			
T	*0*					7	[8]	6		
G	*0*						6	[10]	8	
C	*0*							8	[12]	10
A	*0*								10	[14]

1. The biggest number is **14**, in the bottom-right cell.
2. The path is the boxed cells: 14 ← 12 ← 10 ← 8 ← 6 ← 8 ← 6 ← 4 ← 2.
3. The step from the boxed 8 in row T back to the boxed 6 in row T goes diagonally, and the step from that 6 back to the boxed 8 in row T goes **left**. So the read skips the second **A** of the reference:

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reference:  T A T T A T G C A
read:       T A T T _ T G C A
```

4. 23 cells. The whole table has 72.